Designing an Optimal Disk Management Scheme for a Hypothetical Operating System

Introduction

Mass storage forms the backbone of nonvolatile data storage in modern computer systems. High-performance computing (HPC) environments demand storage solutions with low latency, high throughput, reliability, and efficient space utilization. As secondary storage devices like HDDs and NVM-based SSDs grow in complexity, careful disk management becomes crucial to maximizing system performance and minimizing downtime.

Disk Performance Requirements for HPC

In HPC environments, the disk management system must meet the following requirements:

- **High Throughput**: Ability to transfer large amounts of data rapidly.
- **Low Latency**: Fast seek and transfer times are critical to minimize I/O delays.
- **Reliability**: Systems must tolerate failures through redundancy and error correction.
- **Efficient Space Utilization**: Minimize fragmentation and manage storage logically.
- **Scalability**: Support increasing workloads and larger datasets.
- **Fairness**: Avoid starvation, ensuring timely servicing of I/O requests.

Disk Scheduling Strategy

Selected Scheduling Algorithm: SCAN (Elevator Algorithm)

Reason for Choosing SCAN:

- SCAN moves the disk head from one end of the disk to the other, servicing requests along the way, and then reverses direction.
- Compared to FCFS, SCAN reduces unnecessary long seeks and provides better average response times.
- SCAN avoids starvation problems associated with SSTF, making it suitable for the heavy, continuous loads in HPC.

Why Alternatives Were Not Selected:

- **FCFS**: Although fair, it results in high total head movement and poor performance in heavy loads.
- **SSTF**: Tends to favor nearby requests, causing starvation for requests far from the current head position.
- **C-SCAN**: Provides uniform wait time but slightly increases average seek time because the head must return rapidly without servicing on the way back.
- **Deadline or CFQ**: Although advanced, these complex algorithms are more tuned for mixed general-purpose workloads rather than pure HPC speed demands.

Thus, **SCAN** offers a perfect balance between fairness and throughput for HPC mass storage.

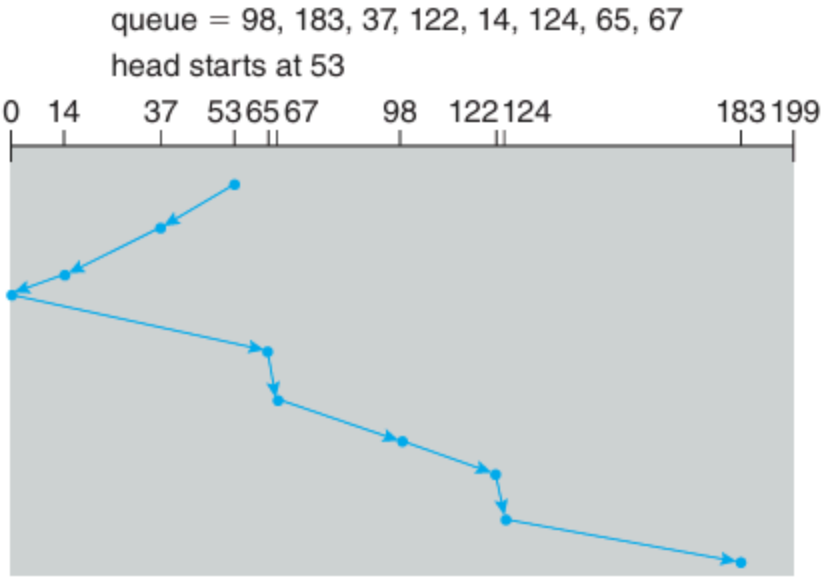


Figure 11.7 SCAN disk scheduling.

Disk Formatting, Space Management, and Data Reliability

Disk Formatting

- **Low-Level Formatting**: Sets up tracks and sectors (for HDDs) or pages and blocks (for NVM devices).
- **Logical Formatting**: Establishes file systems, such as ext4 or XFS, ensuring optimal file access.

Space Management

- **Bitmap-Based Free Space Management**:
    - Efficient for quickly checking free or used sectors or blocks.
    - Suitable for large storage arrays typically used in HPC environments.

- **Extent-Based Allocation**:
    - Allocates contiguous blocks for files, reducing seek time and enhancing sequential read/write performance.

Data Reliability

- **RAID Protection**:
    - RAID-6 or RAID-10 is recommended to tolerate multiple disk failures without data loss.
- **Error Correcting Codes (ECC)**:
    - Both HDDs and SSDs use ECC during read/write operations to detect and correct minor errors automatically.
- **Over-Provisioning in SSDs**:
    - Extra storage set aside for wear leveling and efficient garbage collection, ensuring longer device lifetimes.
- **Snapshots and Backups**:
    - Regular snapshot mechanisms and offsite/cloud backups protect against catastrophic failures.

Justification of Design Decisions

- **SCAN scheduling** minimizes average seek times without starving any request, important for predictable HPC I/O behavior.
- **Bitmaps** offer fast and scalable free space tracking compared to linked-list allocation.

- **Extent allocation** optimizes sequential access patterns common in large-scale simulations and datasets.

- **RAID-6 and RAID-10** provide reliability without sacrificing too much performance, essential for critical computing operations.

- **Over-provisioning and wear leveling** extend SSD lifespan, crucial for minimizing downtime.

Conclusion

Designing a disk management system for an HPC environment requires careful consideration of storage structures, device characteristics, and scheduling algorithms. By combining SCAN disk scheduling, efficient space management techniques, and strong reliability measures like RAID and ECC, the proposed scheme ensures high throughput, low latency, and dependable data storage — all essential for supporting the intensive demands of high-performance computing systems.